

TROUBLE SHOOTING AND HELP DESK

TASK: 1

List 5 common BIOS or POST error messages you might see when starting a PC and write what hardware or configuration issue each one usually points to.

BIOS/POST Error Message	Hardware or Configuration Issue
CMOS Checksum Error – Defaults Loaded	Usually indicates corrupted/reset CMOS settings or a weak CMOS/RTC battery.
Keyboard Error or No Keyboard Present	The keyboard may be disconnected, faulty, or not detected by the system.
CPU Fan Error	The CPU fan may be disconnected, not spinning correctly, or its BIOS fan setting may require checking.
Memory Error / RAM Error	Usually points to incorrectly seated, incompatible, or faulty RAM.
Boot Device Not Found / No Boot Device	The storage drive may not be detected, or the BIOS boot order may be incorrect.

A CMOS checksum error can occur when the system cannot validate its stored CMOS configuration; HP lists a failed system-board battery and an incorrectly updated BIOS among possible causes.

TASK: 2

Given this scenario: On boot, a PC beeps 3 times in a repeating pattern and does not display anything on the screen. Research and write what this POST beep code typically means for an AMI BIOS, and suggest two troubleshooting steps.

Three Beeps on an AMI BIOS

For the **classic AMI BIOS beep-code table**, **3 short beeps** indicate a **Base 64 KB memory failure**, meaning the system has detected a problem with system memory.

Troubleshooting Steps

Step 1 – Reseat the RAM:

Turn off and unplug the PC, remove the RAM module, clean/check the contacts if appropriate, and reinstall it firmly in the recommended DIMM slot.

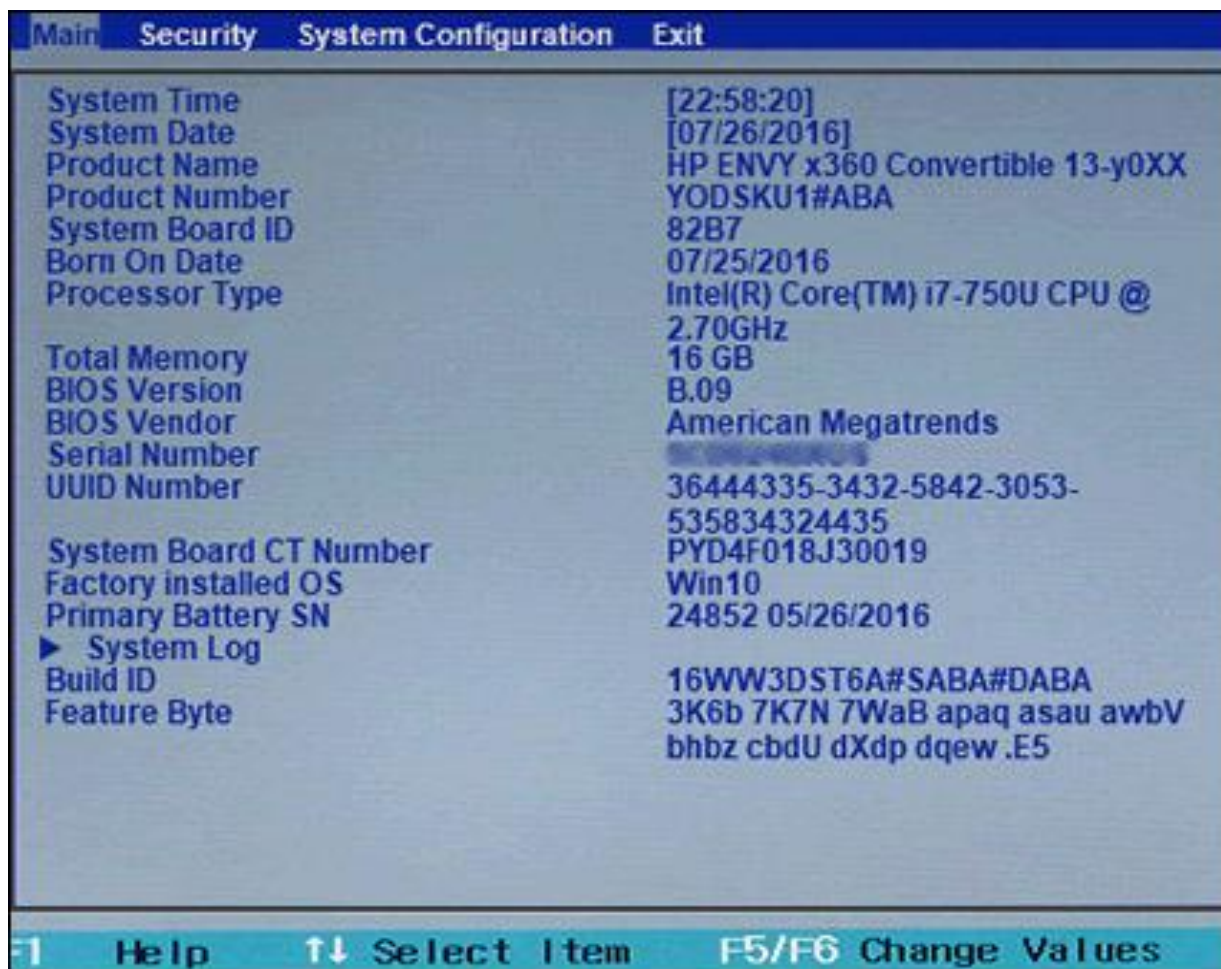
Step 2 – Test the RAM:

If there are multiple RAM modules, test one module at a time and try the recommended slots. If the problem continues, test with known-good compatible RAM.

Important: Beep-code meanings can vary between BIOS/UEFI versions and motherboard manufacturers. For example, ASRock documents **3 long beeps** on a particular AMI UEFI implementation as indicating missing memory, so the motherboard manual should be checked before making a final diagnosis.

TASK: 3

Access the BIOS setup utility on any available PC or virtual machine, take a screenshot of the main BIOS screen, and identify the motherboard manufacturer and BIOS version.



TASK: 4

You are given a screenshot of a POST error message that says 'CMOS checksum error – Defaults loaded.' Explain what this means and describe how you would resolve this issue in a real system.

Hint: Think about the CMOS battery and BIOS settings reset.

CMOS Checksum Error – Defaults Loaded

Meaning

The message “**CMOS checksum error – Defaults loaded**” means that the BIOS detected that the saved CMOS configuration did not pass its checksum/integrity check. As a result, the BIOS has loaded its default configuration instead of using the previously saved settings.

A common cause is a **weak or failed CMOS/RTC battery**, although an incorrectly completed BIOS update or corrupted BIOS/CMOS settings can also cause the problem.

How to Resolve It

1. Restart the PC and enter the **BIOS/UEFI Setup**.
2. Select **Load Setup Defaults / Optimized Defaults**.
3. Check the system **date and time**.
4. Check the **boot order** and make sure the correct operating-system drive is selected.
5. Save the BIOS settings and restart the computer.
6. If the error returns repeatedly, especially if the date and time keep resetting, switch off and unplug the PC and replace the **CMOS/RTC battery** according to the motherboard manufacturer's instructions.
7. Enter BIOS again, set the required configuration, save the settings, and restart.

If the error began immediately after a BIOS update, the BIOS/firmware update should also be investigated using the manufacturer's recovery procedure.

Conclusion

The error does not necessarily mean that the motherboard is permanently damaged. In many cases, restoring the BIOS defaults and replacing a weak CMOS/RTC battery resolves the problem. If the error continues after these steps, the motherboard or BIOS firmware may require further diagnosis.

troubleshoot